

Properties of Fourier Series

Fourier series has the following properties:

- **Fourier Series of a constant:** the Fourier series of a constant signal is only the 0 frequency component, i.e. the average of the signal (since the signal does not change), which is equal to the amplitude of the function itself:
 $s(t) = k \rightarrow c_n = c_0 = k$
- **Linearity:** the coefficients of the linear combination of two periodic signals are equal to the linear combination of the coefficients of the original signals:
 $as_1(t) + bs_2(t) \implies ac_{n1} + bc_{n2}$
- **Delay/Anticipation:** the coefficients of a time-shifted signal are equal to the coefficients of the original signal multiplied by a complex exponential (they are phase shifted):
 $s(t) \rightarrow c_n \implies s(t - t_1) \rightarrow c_n \cdot e^{-j\frac{2\pi nt_1}{T_0}}$
- **Phase-shift:** multiplying a signal by a complex exponential, (with frequency multiple of the fundamental one), results in a frequency shift of the Fourier series coefficients of the new signal:
 $s(t) \rightarrow c_n^{(s)}, \quad v(t) = s(t) \cdot e^{j\frac{2\pi n_1 t}{T_0}} \implies v(t) \rightarrow c_{n-n_1}^{(v)}$
- **Integral (for signals with periodic integral):** the coefficients of the integral over the period of a periodic signal are multiplied by a coefficient inversely proportional to frequency:
 $s(t) \rightarrow c_n \implies \int_0^t s(\tau) d\tau \rightarrow j\frac{T_0}{2\pi n} c_n$
- **Derivative (for all signals):** the coefficients of the derivative over the period of a periodic signal are multiplied by a coefficient directly proportional to frequency (inverse of what happens with integration):
 $s(t) \rightarrow c_n \implies \frac{d}{dt} s(t) \rightarrow j\frac{2\pi n}{T_0} c_n$

Properties of Fourier transform

- **Linearity:** the transforms of the linear combination of two aperiodic signals are equal to the linear combination of the transforms of the original signals:
 $as_1(t) + bs_2(t) \implies aS_1(f) + bS_2(f)$
- **Delay/anticipation:** the transform of a time-shifted signal is equal to the transform of the original signal multiplied by a complex exponential (it is phase-shifted):
 $s(t) \rightarrow S(f) \implies s(t - t_1) \rightarrow S(f) \cdot e^{-j2\pi ft_1}$
- **“Phase-shift” (frequency shift):** multiplying a signal by complex exponential, (with frequency multiple of the fundamental one), results in a frequency shift of the Fourier transform of the new signal
 $s(t) \rightarrow S(f) \implies s(t) \cdot e^{j2\pi f_1 t} \rightarrow S(f - f_1)$
- **Integral:** the transform of the integral over the period of signal are multiplied by a factor of $-j\frac{1}{2\pi f}$, inversely proportional to frequency:
 $s(t) \rightarrow S(f) \implies \int_{-\infty}^t s(t) dt \rightarrow -j\frac{1}{2\pi f} S(f)$
- **Derivative:** the transform of the derivative over time of a signal is multiplied by a complex coefficient directly proportional to frequency (inverse of what happens with integration):
 $s(t) \rightarrow S(f) \implies \frac{d}{dt} s(t) \rightarrow j2\pi f S(f)$